Chapter 23 - Semantic Reasoning: Stage 6 of the Semantic Knowledge Development Lifecycle

Deriving New Knowledge Through Automated Inference

The first five stages of the Semantic Knowledge Development Lifecycle (SKDL) transformed an abstract conceptual vocabulary into a **governed**, **reusable**, **validated** knowledge asset.

- Conceptual Modeling gave us structure.
- Semantic Description gave us meaning.
- Knowledge Reuse gave us composability.
- Semantic Governance gave us discipline.
- Semantic Validation gave us trust.

Now, as knowledge begins to flow from ontology models into production systems, a new challenge emerges:

> **"What new knowledge can we derive from what we already know?"**

Semantic Reasoning answers this question.

It is the discipline of deriving new, logical entailed knowledge from an existing ontology through automated inference.

> It transforms a static knowledge base into a dynamic, intelligent system capable of discovering relationships, classifications, and consequences that were never explicitly asserted.

Reasoning is the engine of intelligence in an ontology.

This chapter examines **Exercise 20** and **Exercise 21** from Michael DeBellis' *Protégé 5 New OWL Pizza Tutorial* through the lens of semantic reasoning. While **Exercise 18** introduced governance mechanisms (disjointness axioms), and **Exercise 19** introduced validation (the Probe Class patter), **Exercise 20** and **Exercise 21** introduce the core distinction that makes reasoning possible:

- **Primitive Classes** (defined with `SubClassOf`) express only necessary conditions. They describe what something *must* be, but they do not enable auto-classification.

- **Defined Classes** (defined with `EquivalentTo`) express both necessary and sufficient conditions. They describe what something *is*, and they also enable the reasoner to classify individuals and classes automatically.

This distinction is the **gateway to automated reasoning**. Once a class is defined with sufficient conditions, the reasoner can recognize instances of that class without being told.

Along the way, we will examine reasoning from multiple complementary perspectives: mathematical, historical, engineering, architectural, and industrial -- revealing that

> **automated reasoning is not a modern invention, but the culmination of more than two thousand years of logical thought.**

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23.1 Chapter Introduction -- From Verification to Discovery

23.1.1 The Journey So Far

The first five stages of the Semantic Knowledge Development Lifecycle (SKDL) established the foundations of ontology engineering.

Stage 1 -- Conceptual Modeling identified the concepts that define a knowledge domain and organized them into a coherent semantic taxonomy.

Stage 2 -- Semantic Description enriched those concepts with formal meaning using Description Logic, OWL restrictions, and class expressions.

Stage 3 -- Knowledge Reuse transformed validated semantic knowledge into reusable assets that could be shared, extended, and specialized systematically.

Stage 4 -- Semantic Governance established the policies, standards, and processes required to maintain semantic integrity across an evolving knowledge ecosystem.

Stage 5 -- Semantic Validation verified that the ontology is logically consistent, semantically complete, and fit for purpose before deployment.

By the end of Stage 5, the `Pizza` ontology had been validated. It was consistent, satisfiable, and free of contradictions. The governance rules had been enforced. The probe class had been created, used, and removed.

The ontology was correct.

But correctness is not the same as intelligence.

A correct ontology that cannot derive new knowledge is like a well-organized library where no noe can find a needed book. It contains knowledge, but it does not *use* knowledge. It is passive, not active.

This is where **Stage 6 -- Semantic Reasoning** enters the lifecycle.

23.1.2 The New Question

Validation asked: **"Is this model broken?**

Reasoning asks: **"What else can this model tell us?"**

The distinction is fundamental.

Validation ensures that the ontology is logically sound. Reasoning derives new knowledge from that sound foundation.

Without validation, reasoning produces unsound inferences.

Without reasoning, validation produces a correct but static model.

Together, they produce trustworthy intelligence.

23.1.3 The Central Insight

The central insight of this chapter is deceptively simple:

> **Reasoning does not generate new facts -- it makes explicit what was already implicit in the axioms.**

The reasoner does not invent knowledge. It discovers the logical consequences of the axioms the engineer has written. Every inferred statement was already entailed by the ontology. The reasoner simply makes the entailment explicit.

This is the power of formal semantic:

> from a seed of **asserted knowledge**, the reasoner grows a tree of **inferred knowledge**.

23.2 Learning Objectives

After completing this chapter, you should be able to achieve the following learning outcomes:

Knowledge:

- Explain why Semantic Reasoning is the sixty stage of the Semantic Knowledge Development Lifecycle (SKDL)
- Distinguish between **primitive classes** (`SubClassOf` only) and **defined classes** (`EquivalentTo`)
- Understand the difference between **asserted knowledge** and **inferred knowledge**
- Explain how a reasoner derives new subclass relationships automatically
- Identify major RDF reasoners beyond Protégé and explain their differences
- Understand how reasoning applies in industrial contexts (problem-solving, decision-making, safety, compliance)
- Explain the mathematical foundations of reasoning: subsumption, entailment, tableau algorithm, decidability

Practical Skills:

- Create a primitive class (`CheesyPizza`) using `SubClassOf` only
- Convert a primitive class to a defined class using `Edit > Convert to defined class`
- Observe the reasoner automatically classifying individuals and classes
- Use Protégé's explanation feature to understand why an inference was made
- Compare reasoning results across different reasoners (Pellet, HermiT, ELK, JFact)
- Select the appropriate reasoner for a given task

Engineering Perspective:

- Understand reasoning as "automated knowledge discovery"
- Explain why defined classes enable automation while primitive classes do not
- Map reasoning activities to the EKA R layer
- Apply reasoning to industrial problem-solving, safety, and compliance scenarios

Mathematical Perspective:

- Understand model-theoretic semantics: interpretations, models, entailment
- Explain subsumption as set inclusion and logical consequence
- Understand the tableau algorithm as a decision procedure
- Explain decidability and computational complexity
- Connect reasoning to Description Logic semantics

Historical Perspective:

- Trace the reasoning tradition from Aristotle's syllogism to modern automated reasoning
- Connect modern OWL reasoners to th 2,000-year quest for automated inference

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